PTX2CT-GAN: Reconstruction of Chest CT from Single X-ray Radiography for Pneumothorax

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***Abstract*—Pneumothorax (PTX) is a common acute condition in thoracic surgery and respiratory medicine, and the accurate diagnosis and quantitative evaluation rely on high-quality three-dimensional (3D) chest imaging that involves high radiation exposure and costs. In this study, we propose PTX2CT-GAN, a generative adversarial learning-based framework that reconstructs 3D chest CT images from a single X-ray of pneumothorax. The framework integrates a Reconstruction Feature Enhancement Module (RFEM) to improve feature extraction and structural representation during CT reconstruction. Experiments on the MY-PTX-998 dataset demonstrate that the proposed PTX2CT-GAN outperforms related methods both on quantitative and qualitative evaluations. This work provides a low-radiation, low-cost, and high-quality alternative for CT imaging in pneumothorax diagnosis, especially in resource-limited clinical environments.**

***Index Terms*—Medical Imaging, Pneumothorax X-ray, Computed Tomography, Reconstruction**

I. Introduction

Pneumothorax (PTX) is one of the most common emergencies in thoracic surgery and respiratory medicine. Its main characteristic is the entry of air into the pleural cavity, resulting in air accumulation that compresses the lung tissue, causing lung collapse and preventing normal expansion. Patients often present with symptoms such as chest pain, chest tightness, and shortness of breath; in severe cases, respiratory failure or even coma may occur. It is estimated that the annual incidence of spontaneous pneumothorax is approximately 22.7 cases per 100,000 people, with a recurrence rate of up to 48% after conservative chest tube treatment [1].

In clinical practice, X-ray and computed tomography (CT) are the primary imaging modalities for diagnosing pneumothorax. Due to its simplicity, low radiation dose, and low cost, X-ray examination is widely used in medical institutions. However, its two-dimensional imaging nature causes overlapping of anatomical structures, which limits the ability to display three-dimensional information adequately, increases the difficulty of image interpretation, and may lead to misdiagnosis.

In contrast, CT scanning is considered the "gold standard" for diagnosing pneumothorax, as it provides high-resolution three-dimensional images that effectively eliminate tissue overlap and perform excellently in determining the presence, extent, and location of pneumothorax. Nevertheless, CT scanning has certain limitations: (1) its relatively high radiation dose poses potential health risks, particularly to sensitive populations such as pregnant women and children; (2) its high cost limits accessibility in regions with scarce medical resources; and (3) in intensive care units (ICUs), patients' unstable conditions and the difficulty of transportation often make CT examinations impractical.

Given these challenges, an important clinical question arises: is it possible to reconstruct chest CT data from a single pneumothorax X-ray, thereby generating high-quality three-dimensional images while reducing radiation exposure? Such an approach could not only provide more accurate and objective quantitative assessments of pneumothorax (e.g., pneumothorax volume and lung compression ratio) but also offer a low-radiation, low-cost imaging alternative in settings with limited medical resources or where CT scans cannot be performed.

In recent years, with the rapid development of artificial intelligence technologies, deep learning algorithms have been proven capable of learning complex nonlinear mappings between two-dimensional X-rays and three-dimensional CT volumetric data. Although several studies [2]–[4] have achieved promising results in X-ray-to-CT reconstruction tasks, most existing methods suffer from insufficient extraction of critical structural features, leading to suboptimal detail restoration and limited structural fidelity in reconstructed results.

Therefore, this paper proposes a medical image reconstruction model named PTX2CT-GAN (PTX in Xay to CT GAN). The model incorporates a Reconstruction Feature Enhancement Module (RFEM) to strengthen the representation of key features during the CT reconstruction process, thereby enabling high-precision chest CT reconstruction from a single pneumothorax X-ray. Our main contributions are summarized as follows:

- We propose a Reconstruction Feature Enhancement Module (RFEM) to enhance the model's feature extraction capability in pneumothorax CT reconstruction.
- We constructed the first paired pneumothorax X-ray and CT dataset, MY-PTX-998, providing a data foundation for cross-modal reconstruction from X-ray to CT images.
- Experimental results on the MY-PTX-998 pneumothorax CT dataset demonstrate that the proposed PTX2CT-GAN achieves superior performance compared to exist-

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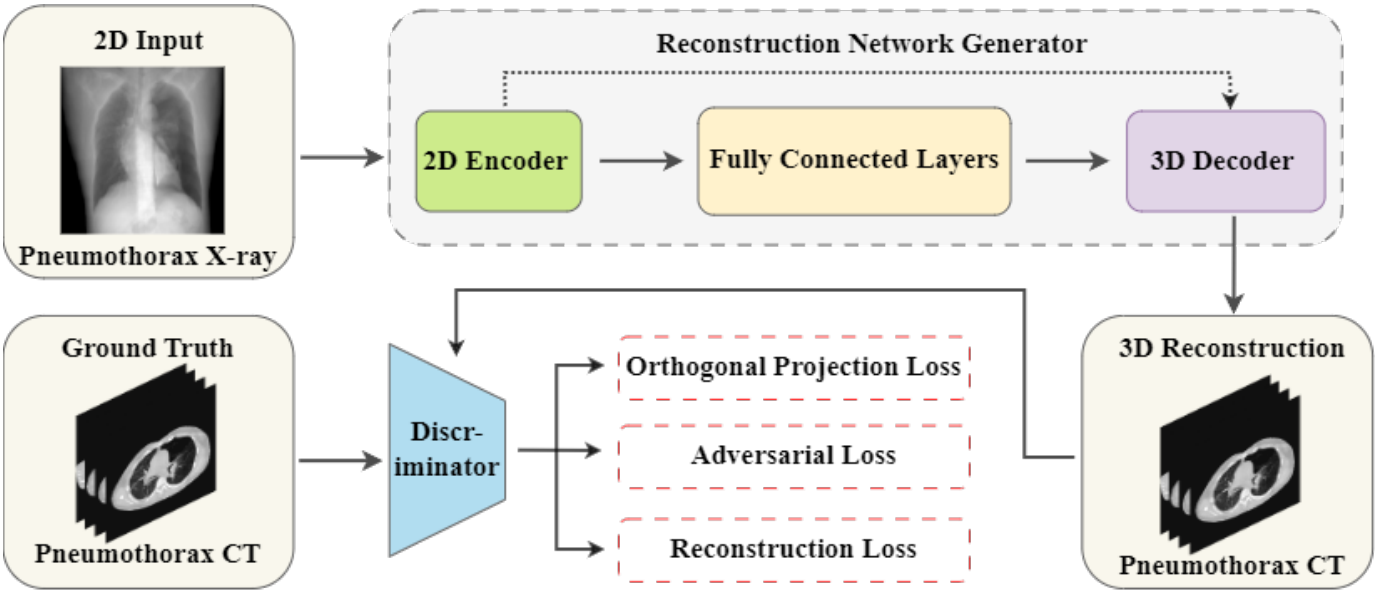

Fig. 1. Overview of the PTX2CT-GAN architecture. The input is a 256×256 pneumothorax X-ray image, which is first processed by a 2D encoder equipped with the Reconstruction Feature Enhancement Module (see Section II-A) to extract and encode features. Subsequently, the encoded features are upsampled and reconstructed through a 3D decoder, ultimately generating a $256 \times 256 \times 256$ three-dimensional pneumothorax CT image.

ing methods in both quantitative metrics and qualitative evaluations

II. Methods

In this section, we present our proposed neural network model architecture PTX2CT-GAN, as shown in Fig. 1, which is designed to reconstruct chest CT images from a single Pneumothorax X-ray.

A. Generator

1) Reconstruction Feature Enhancement Module: We designed a Reconstruction Feature Enhancement Module (RFEM), as shown in Fig. 2. The RFEM assigns attention weights to the features extracted by the Dense Block at each encoding stage across both channel and spatial dimensions, enhancing critical features while suppressing less important ones. This approach more effectively captures key anatomical structures in CT reconstruction, allowing the model to focus on important areas of the image and ultimately improving reconstruction accuracy.

Specifically, for the input feature map F_{input} to the RFEM, it is first downsampled through a convolutional layer. Next, the feature map enters the Dense Block, where dense connections and feature reuse mechanisms ensure that the output feature map F_{dense} effectively retains low-level features and transmits them to higher levels. In the task of reconstructing 3D CT images from a pneumothorax X-ray, low-level features often contain important details that are crucial for constructing higher-level global features. In this way, the RFEM enhances the correlation between features at different levels, with a particular emphasis on the importance of low-level features, which play a critical role in capturing subtle structures and details during the reconstruction process. The representation of F_{dense} is as follows:

$$F_{\text{dense}} = \text{Concat}(F_1, F_2, \ldots, F_N) \tag{1}$$

where F_{N} represents the output feature map of the N_{th} dense layer in the encoding stage, and Concat refers to concatenating the output feature maps of each dense layer along the channel dimension.

To enhance important features in CT reconstruction and suppress less important ones, we applied a channel attention mechanism to F_{dense}. Specifically, F_{dense} is first transformed into two feature vectors through Global Average Pooling (GAP) and Global Max Pooling (GMP). These two feature vectors are then processed through a shared fully connected layer and added together, followed by a Sigmoid activation function to generate the final channel attention weights W_{ch} .The representation of W_{ch} is as follows:

$$\begin{aligned} W_{\text{ch}} = \sigma\big(&\text{ReLU}\big(\text{FC}\big(\text{GAP}(F_{\text{dense}})\big)\big) \\ &+ \text{ReLU}\big(\text{FC}\big(\text{GMP}(F_{\text{dense}})\big)\big)\big) \end{aligned} \tag{2}$$

where σ represents the Sigmoid activation function, and the fully connected layer is represented by FC.

Then, the channel attention weights W_{ch} are then applied to the feature map through element-wise multiplication.

$$F_{\text{ca}} = F_{\text{dense}} \odot W_{\text{ch}} \tag{3}$$

Next, global spatial information W_{sp} is extracted from the channel-weighted feature map F_{ca}, and then applied to F_{ca}, resulting in the spatially weighted feature map F_{sa}. The equation is as follows:

$$W_{\text{sp}} = \sigma\left(\text{Conv}_{7\times7}\left([\text{GAP}(F_{\text{ca}}); \text{GMP}(F_{\text{ca}})]\right)\right) \tag{4}$$

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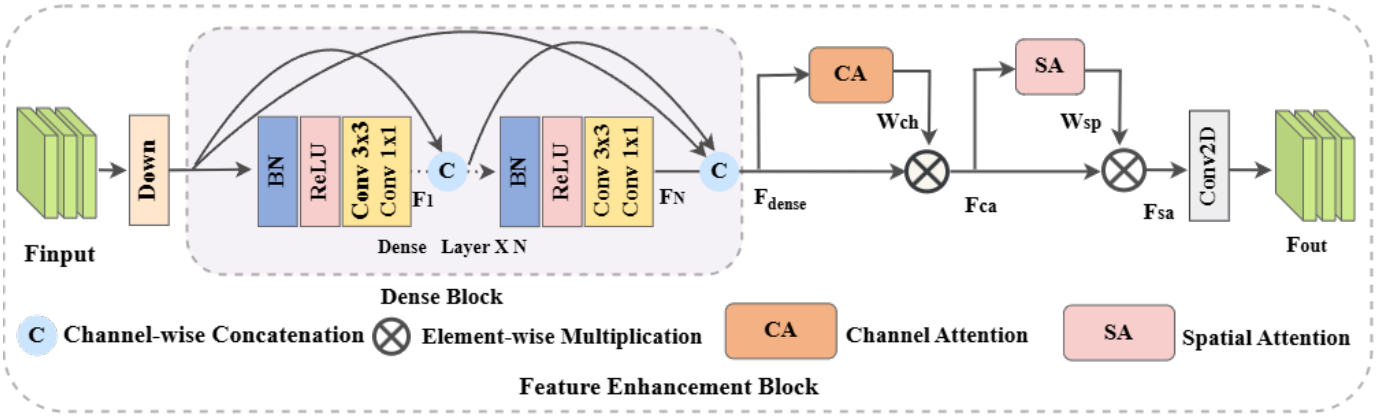

Fig. 2. Illustration of the proposed Reconstruction Feature Enhancement Module. The channel information W_{ch} is extracted from the feature map F_{dense}, and the spatial information W_{sp} is extracted from the feature map F_{ca}.

$$F_{\text{sa}} = F_{\text{ca}} \odot W_{\text{sp}} \tag{5}$$

Finally, F_{sa} is compressed through a convolutional layer, reducing the number of output channels to half of the input feature map's channels, thereby generating the final output feature map F_{output}.

$$F_{\text{output}} = \text{Conv2D}(F_{\text{sa}}) \tag{6}$$

B. Discriminator

Given the complexity of generating chest CT images from a single pneumothorax X-ray, this study adopts a 3D Patch Discriminator [5], following previous works, to replace the traditional GAN discriminator. Unlike approaches that only discriminate individual CT slices, the proposed discriminator takes 3D image patches containing spatial and volumetric information as input, enabling it to more effectively distinguish real CT images from generated ones.

C. Loss Function

During the training process, the network model continuously optimizes its parameters by minimizing a specific loss function. Therefore, the total loss functions for the discriminator (D) and generator (G) are defined as follows.

$$D^* = \arg\min_D \lambda_1 \mathcal{L}_{\text{LSGAN}}(D) \tag{7}$$

$$G^* = \arg\min_G \left[\lambda_1 \mathcal{L}_{\text{LSGAN}}(G) + \lambda_2 \mathcal{L}_{\text{re}} + \lambda_3 \mathcal{L}_{\text{op}}\right] \tag{8}$$

where $\mathcal{L}_{\text{LSGAN}}$, is adversarial loss [6], $\mathcal{L}_{\text{re}}$ is reconstruction loss [4] and $\mathcal{L}_{\text{op}}$ is orthogonal projection loss. λ_1, λ_2, and λ_3 are weights used to control the relative importance of each loss component in the total loss. In this experiment, $\lambda_1 = 0.1$, $\lambda_2 = 1$, and $\lambda_3 = 10$. The $\mathcal{L}_{\text{LSGAN}}$ is defined as follows:

$$\begin{aligned}\mathcal{L}_{\text{LSGAN}}(D) = \frac{1}{2}\Big[&\mathbb{E}_{y\sim p_{(\text{CT})}}\left(D(y|x)-1\right)^2 \\ &+\mathbb{E}_{x\sim p_{(X-ray)}}\left(D(G(x)|x)-0\right)^2\Big]\end{aligned} \tag{9}$$

$$\mathcal{L}_{\text{LSGAN}}(G) = \frac{1}{2}\left[\mathbb{E}_{x\sim p_{(X-ray)}}\left(D(G(x)|x)-1\right)^2\right] \tag{10}$$

where x is the input pneumothorax X-ray, and y is the reconstructed pneumothorax CT. The discriminator D and the generator G are trained alternately to compete with each other, p represents the corresponding data distribution, and $\mathbb{E}$ represents the maximum likelihood estimate.

The $\mathcal{L}_{\text{re}}$ is defined as follows:

$$\mathcal{L}_{\text{re}} = \mathbb{E}_{x,y}\left\|y - G(x)\right\|_2^2 \tag{11}$$

where y denotes the ground truth CT, and $G(x)$ represents the CT reconstructed by the generator.

The $\mathcal{L}_{\text{op}}$ is defined as follows:

$$\mathcal{L}_{op} = \frac{1}{3}\sum_{i\in\{ax,co,sa\}} \mathbb{E}_{x,y}\|P_i(y) - P_i(G(x))\|_1 \tag{12}$$

where ax, co, and sa represent the projections in the axial, coronal, and sagittal planes, respectively. $P_i(y)$ represents the projection result of the real CT image y on projection plane i, and $P_i(G(x))$ represents the projection result of the generated CT image $G(x)$ on projection plane i.

III. Experiments And Results

A. Dataset and Preprocessing

To train and validate the reconstructed CT images generated by the PTX2CT-GAN model, it is necessary to build an X-CT paired dataset consisting of pneumothorax X-ray and CT data from the same patient. However, due to the self-absorbing nature of pneumothorax, it is difficult to obtain paired imaging data from patients who undergo both X-ray and CT examinations within a very short period of time (within 24 hours).

To address this issue, we collected 998 pneumothorax CT cases from Mianyang Central Hospital, covering the period from 2013 to 2019, and employed Digital Reconstructed Radiography (DRR) technology [7], as shown in Fig. 3 to generate corresponding two-dimensional pneumothorax X-ray images from real CT volume data. This process resulted in the construction of a pneumothorax X-ray and CT paired

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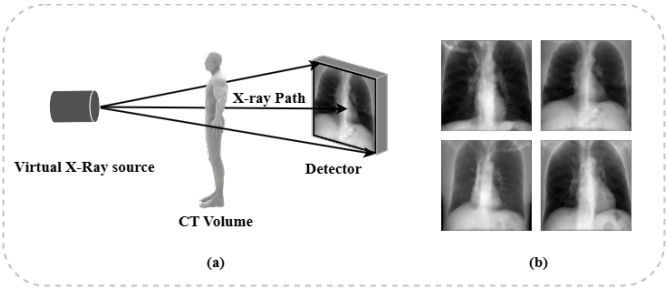

Fig. 3. (a) Schematic diagram of obtaining DRR images from 3D CT volumes; (b) pneumothorax X-ray image obtained using DRR technology from the MY-PTX-998 dataset.

dataset named MY-PTX-998, in which 900 cases were used for training and 98 cases for testing.

In addition, image preprocessing was performed to effectively remove irrelevant noise in the CT images (such as examination tables and patient clothing), as shown in Fig. 4. All 998 CT scans were then resampled to an isotropic voxel resolution of $1 \times 1 \times 1$ mm^3 and centrally cropped to a size of $256 \times 256 \times 256$ to ensure stable model convergence and reduce computational cost.

B. Implementation Details

We used the Adam optimizer to train the network model, with an initial learning rate set to 2e-4 and momentum parameters β_1 and β_2 set to 0.5 and 0.99, respectively. The entire model training process consisted of 100 epochs. During the first 70 epochs, the initial learning rate was kept constant. In the subsequent 30 epochs, we employed a linear learning rate decay strategy, gradually reducing the learning rate to zero. All experiments were conducted on an NVIDIA RTX A100 GPU (with 80 GB of memory) for both training and testing, implemented using the PyTorch framework. Due to GPU memory constraints, the resolution of a single 2D pneumothorax X-ray input to the model was set to 256x256, and the batch size for all experiments was set to 4.

C. Evaluation Metrics

To evaluate the performance of PTX2CT-GAN in reconstructing CT images, we employed four common medical image reconstruction metrics: Peak Signal-to-Noise Ratio (PSNR), in Equation 13, Structural Similarity Index (SSIM) [8], in Equation 14, Learned Perceptual Image Patch Similarity (LPIPS) [9].

$$\text{PSNR}(I_{\text{gt}}, I_{\text{re}}) = 10 \cdot \log_{10}\left(\frac{\text{MAX}^2}{\text{MSE}(I_{\text{gt}}, I_{\text{re}})}\right) \tag{13}$$

where I_{gt} and I_{re} represent the real pneumothorax CT slice and the reconstructed pneumothorax CT slice, respectively, MSE represents the mean squared error, and MAX is the maximum pixel value of the slice.

$$\text{SSIM}(I_{\text{gt}}, I_{\text{re}}) = \frac{(2\mu_{I_{\text{gt}}}\mu_{I_{\text{re}}} + c_1)(2\sigma_{I_{\text{gt}},I_{\text{re}}} + c_2)}{(\mu_{I_{\text{gt}}}^2 + \mu_{I_{\text{re}}}^2 + c_1)(\sigma_{I_{\text{gt}}}^2 + \sigma_{I_{\text{re}}}^2 + c_2)} \tag{14}$$

where, μ is the mean, σ is the variance, σ_{xy} is the covariance, and the constant c is used to avoid division by zero.

D. Quantitative Results

We compared our method with several existing approaches, including 2DCNN [3], ReconNet [2], X2CT-GAN [4], DP-GAN [10], AECT-GAN [11], and X-Recon [12]. Table I summarizes the quantitative experimental results of these methods. It is important to note that all quantitative metrics were evaluated based on 256 slices each from the axial, coronal, and sagittal planes of the CT volume images in the test set, and the average score for the entire CT.

TABLE I
QUANTITATIVE COMPARISON OF PTX2CT-GAN AND OTHER METHODS ON THE TEST SET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND "+S" INDICATES THAT THE INPUT FOR THAT METHOD IS A SINGLE PNEUMOTHORAX CHEST X-RAY.

Methods	PSNR(dB)↑	SSIM(%)↑	LPIPS↓
ReconNet [2]	44.45	54.98	39.91
2DCNN [3]	51.16	75.65	26.35
X2CT-GAN+S [4]	50.02	73.93	20.44
DP-GAN+S [10]	50.25	74.94	20.40
X-Recon+S [12]	50.17	74.51	19.94
AECT-GAN+S [11]	50.50	75.12	18.32
PTX2CT-GAN(**Ours**)	**51.56**	**78.47**	**16.14**

Quantitative results show that our proposed method has significant improvements over the comparison methods in terms of three key image quality evaluation indicators: PSNR, SSIM, and LPIPS. Specifically, compared with X2CT-GAN [4], our method improves PSNR by 1.54 dB, SSIM by 4.54, and reduces LPIPS by 4.30. For traditional convolutional neural network methods (such as ReconNet [2] and 2DCNN [3]) and the depthwise separable convolution-based DP-GAN [10], our method achieves better performance in the LPIPS metric, reducing it by 23.77, 10.21, and 4.26, respectively.

E. Qualitative Results

To more intuitively demonstrate the quality of CT images reconstructed from a single pneumothorax X-ray, Fig. 5 compares the axial CT reconstruction results obtained by different methods. It can be observed that the CT images reconstructed by CNN-based methods such as ReconNet [2] and 2DCNN [3] are generally blurry and fail to clearly present thoracic structures. In contrast, compared with other GAN-based methods, the proposed approach is capable of generating higher-quality CT images and more accurately reconstructing pneumothorax regions (as highlighted by the red boxes in the figure).

IV. CONCLUSION

In this paper, we propose a novel deep learning model, PTX2CT-GAN, which reconstructs high-quality three-dimensional (3D) chest computed tomography (CT) images

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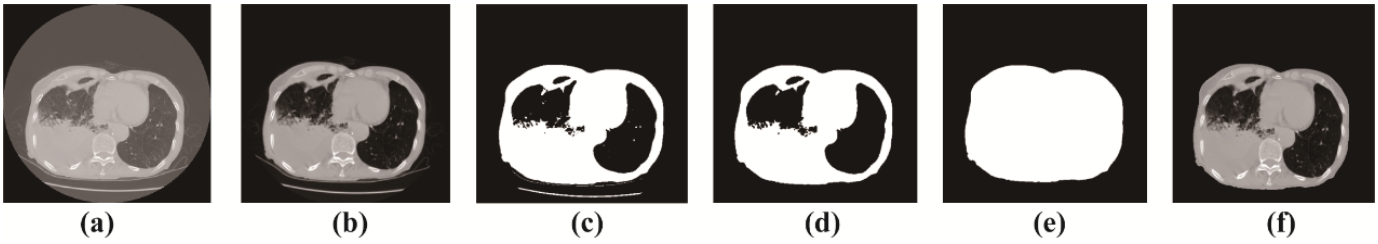

Fig. 4. Pneumothorax CT data preprocessing. (a) Original pneumothorax CT slice image; (b) pixel normalization to the range [0, 255]; (c) thresholding to obtain a binary image; (d) largest connected region in the binary image; (e) flood fill operation applied to the largest connected region; (f) extraction of the largest connected region.

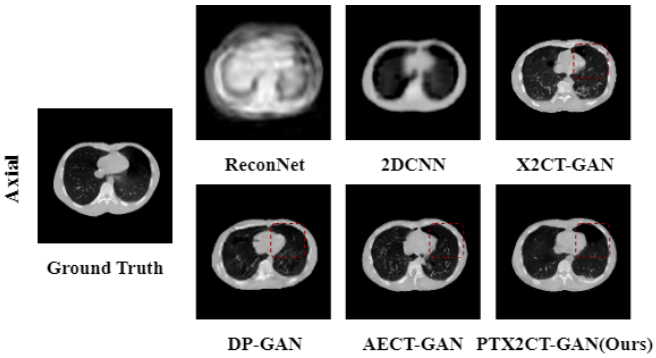

Fig. 5. CT volumetric image slices were reconstructed from a single pneumothorax X-ray using different methods.

from a single pneumothorax X-ray. Unlike traditional analytical reconstruction, iterative reconstruction methods, and dual-view deep learning models that typically require multiple X-ray projections as input, our approach can directly infer a complete 3D volumetric CT structure from a single two-dimensional (2D) pneumothorax X-ray image.

To enhance the model's capability in feature extraction during pneumothorax CT reconstruction, we design a Reconstruction Feature Enhancement Module (RFEM) within the encoder. Experimental results on clinical pneumothorax datasets demonstrate that PTX2CT-GAN significantly outperforms existing state-of-the-art reconstruction methods in terms of quantitative metrics (such as PSNR and SSIM) and qualitative visual evaluation, verifying its effectiveness and potential for pneumothorax CT reconstruction tasks.

Ethical Approve

The collection and use of all pneumothorax data strictly followed the ethical standards and regulatory guidelines of Mianyang Central Hospital. All patient information was anonymized prior to data analysis to ensure privacy and confidentiality. The study protocol was reviewed and approved by the Biomedical Ethics Committee of Mianyang Central Hospital (ethical approval number: S20230377-01).

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